

Muhammad Ahtesham

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Summary

Production-focused AI Engineer with hands-on experience building LLM-integrated pipelines, AI agent workflows, RAG systems, and generative AI applications. Proficient in prompt engineering (chain-of-thought, few-shot, ReAct), LangChain, FastAPI, and Python with a track record of deploying AI APIs serving real production clients. Complementary full-stack background in Node.js and React.js enables end-to-end ownership of AI-powered features. Seeking an AI Engineer / ML Engineer role to build intelligent, production-ready AI systems at scale.

Technical Skills

AI & LLM: LLM Integration, RAG (Retrieval-Augmented Generation), Prompt Engineering, AI Agent Development, Fine-tuning, Chain-of-Thought, Few-Shot Learning, ReAct Prompting, Function Calling

Generative AI: GPT-4, GPT-3.5, Gemini API, OpenVoice V2, Zero-Shot Synthesis, Embeddings, Vector Databases (FAISS, Pinecone), Semantic Search

ML & Tooling: LangChain, LangGraph, n8n, Hugging Face, OpenAI API, Google Generative AI SDK, Structured Output Parsing

Backend & Cloud: FastAPI, Node.js, Express.js, RESTful APIs, GraphQL, Python, Microservices, AWS (EC2, S3, Lambda), GCP, Azure, Docker, CI/CD, GitHub Actions

Languages: Python, JavaScript, TypeScript, SQL, Java, C++ **Databases:** PostgreSQL, MongoDB, MySQL, Firebase, Supabase, Redis

Testing & Practices: Jest, JUnit, Mocha, Pytest, TDD, Agile, Scrum, Git Flow, System Design, Scalable AI Architecture

Education

FAST National University of Computing and Emerging Sciences

Aug 2022 – June 2026

Bachelor of Science in Software Engineering *Coursework: Machine Learning, Artificial Intelligence, Data Structures & Algorithms, Database Systems, Software Architecture*

Experience

AI Engineer

Zevenz

Oct 2025 – Mar 2026

- Designed and deployed **3 production LLM-integrated AI agent workflows** using Python and FastAPI, reducing manual processing time by **60%**; engineered prompt pipelines for GPT-4 and Gemini APIs using chain-of-thought, few-shot, and ReAct techniques, improving structured output accuracy by **35%**
- Architected a RAG pattern using vector embeddings and semantic search to ground LLM responses in domain-specific knowledge, significantly reducing hallucination rate
- Orchestrated multi-step AI workflows with n8n, saving **8+ hrs/week**; took full ownership of backend API design and deployment within a 3-person startup, delivering all features on time

Full Stack Engineer

App Minds Software House

Jun 2025 – Aug 2025

- Built and shipped **5+ production features** using React.js and Node.js, reducing frontend load time by **30%** via code splitting and lazy loading; designed RESTful APIs improving data retrieval latency by **25%**
- Collaborated in a 6-person Agile team with Git Flow and PR reviews, maintaining **95%+** sprint velocity consistently

Backend Developer

Nexium Software House (Remote)

Jun 2025 – Aug 2025

- Developed **10+ RESTful API endpoints** in Node.js/Express.js; optimized PostgreSQL via composite indexing and connection pooling, reducing execution time by **40%** and enabling **3×** higher concurrent load
- Implemented JWT auth middleware and RBAC; achieved **80%+** test coverage with Jest TDD integrated in CI/CD pipeline

Projects

Generative AI Voice Cloning Pipeline

OpenVoice V2, FastAPI, Python, n8n

GitHub

- Built end-to-end generative AI pipeline for **zero-shot voice cloning** using OpenVoice V2 — no target speaker training data required; orchestrated with n8n reducing manual steps by **70%**; exposed as versioned FastAPI REST endpoint with Docker-based horizontal scalability

Stratify – AI Powered Trading SaaS

Gemini Flash API, LLM Pipeline, Node.js, Angular, MongoDB

GitHub

- Designed a **RAG-style LLM pipeline** using Gemini Flash API to parse natural language trading strategies into structured JSON at **90%+** accuracy; employed prompt chaining and output validation guardrails for type-safe structured outputs
- Implemented backtesting engine on 2+ years of historical market data with results visualization in the Angular SPA

Blockchain Attendance Management System

Custom Blockchain, SHA-256, Proof-of-Work

GitHub

- Built full-stack blockchain from scratch using custom SHA-256 hashing and Proof-of-Work to store **500+ records** immutably (no third-party libs); O(n) chain validation with tamper-proof integrity; REST API with RBAC

Zevenz Builders Marketplace

React.js, Node.js, MongoDB, Socket.io, Nodemailer

GitHub

- Built production MERN marketplace with real-time Socket.io messaging (**sub-200ms latency**) and **100+ daily** automated email notifications via Nodemailer

Arabic Text Editor

Java, Spring Boot, JUnit, Abstract Factory, DAO

GitHub

- NLP-adjacent Java desktop app with English-to-Arabic conversion using three-layered architecture; **85%+** JUnit test coverage across all modules